

Benchmarks Hide It: Quantization Silently Degrades Spatial Grounding in Edge Vision-Language Models

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Abstract

Vision-language models (VLMs) reach phones, drones, and robots almost exclusively through post-training quantization, and the benchmarks used to certify those compressed checkpoints—VQA, hallucination, OCR, reasoning—report them as near-lossless. We show that this certification is blind to a systematic failure mode. Quantizing Qwen3-VL-2B to 4 bits costs only 0.6 points of POPE and 3.1 points of VQAv2, but degrades spatial grounding in a way whose measured size depends entirely on the metric chosen: on full RefCOCO test splits the standard threshold metric (REC@0.5) reports -3.8 points while a precision-sensitive metric (mean accuracy over IoU 0.50:0.95) reports -6.6 , and on held-out open-world detection the loss concentrates on small objects. The vulnerability is not diffuse. A forward-only probe that promotes one module at a time localizes grounding sensitivity to a contiguous band of mid-network attention modules, statistically unrelated to the module ranking obtained from a VQA probe (Spearman $\rho = 0.001$, top-7 overlap 0/7) and specific to the deployed quantizer ($\rho = 0.08$ between round-to-nearest and GPTQ maps). Keeping only that band at 8 bits—+6% of decoder weight memory, no retraining, no calibration change—recovers a third to a half of the loss under every metric, beats random and VQA-guided allocation at identical memory, and leaves VQAv2 and POPE unharmed. We argue that capability-specific evaluation, not aggregate accuracy, is the reliability requirement for compressed VLMs, and release the audit harness that produces these numbers.

Introduction

A compressed model is certified by the benchmarks its release reports. For vision-language models, those benchmarks are overwhelmingly language-shaped: every major VLM post-training-quantization (PTQ) method—MBQ (Li et al. 2025), Q-VLM (Wang et al. 2024), VLMQ (Xue et al. 2025), LUQ (Bhatnagar et al. 2025), SPEED-Q (Guo et al. 2025)—validates on VQA, OCR, chart, document, and reasoning suites. We verified by full-text search that none of these papers reports a localization metric; SPEED-Q does so while explicitly targeting smartphone and robot deployment. Meanwhile the deployment path is real and quantized by default: the default tags of a popular local-inference distribution of Qwen2.5-VL serve 4-bit weights on a page that advertises bounding-box localization as a feature (Ollama 2026).

That combination is a reliability hazard: the capability edge deployment usually exists for—knowing *where* something is—is the capability no one measures after compression. It is not a hypothetical concern. In the neighbouring literature on visual-token pruning, localization is known to collapse long before question answering does (Endo, Wang, and Yeung-Levy 2025; Huang et al. 2026).

This paper asks what quantization actually does to grounding, and what a practitioner would have to measure to notice. Our contributions are:

1. **A failure mode that standard evaluation misses.** At 4 bits, aggregate hallucination and VQA scores move by 0.6 and 3.1 points, while grounding degrades far more—and the size of the reported degradation depends on the metric: a precision-sensitive IoU average reveals $1.5\text{--}1.8\times$ the damage that the conventional REC@0.5 reports (Figure 1a). At 3 bits under GPTQ the separation becomes categorical: grounding output collapses to zero parseable boxes while VQA retains partial function.
2. **A diagnosis: the vulnerability is capability-specific and localized.** A promote-one-module probe finds grounding sensitivity concentrated in a mid-network attention band, uncorrelated with the VQA sensitivity ranking and specific to the deployed quantizer (Figure 1b).
3. **A cheap, training-free mitigation and an audit protocol.** Protecting the band at 8 bits costs +6% decoder weight memory and recovers a third to a half of the loss on full official splits with image-clustered confidence intervals, while matched-memory random and VQA-guided allocation do not.

Setup

Model and quantization. Qwen3-VL-2B-Instruct (Bai et al. 2025), which emits boxes as text in a normalized $[0, 1000]$ coordinate system, is edge-realistic (~ 1.5 GB at W4). Following (Shin et al. 2026) the vision tower is left at BF16 and all aggressive quantization targets the language decoder. We use simulated round-to-nearest (RTN, groupwise symmetric, group 128) and our own GPTQ implementation with wikitext calibration, at 8/4/3 bits, plus a W4A8 stress condition with per-token dynamic 8-bit activations. We write W_xA_y for x -bit weights and y -bit activations.

Table 1: Capability retention under quantization (absolute scores, identical pipeline every row). Aggregate language-shaped benchmarks stay flat where grounding degrades; at 3 bits the capabilities separate outright.

Setting	Bits	REC@0.5	VQAv2	POPE
BF16 reference	16	88.9	81.1	89.4
RTN	8	88.7	81.3	89.4
RTN	4	84.5	78.0	88.8
GPTQ	4	85.4	78.1	89.4
RTN (W4A8)	4	82.8	77.5	88.7
RTN	3	0.0	1.3	50.4
GPTQ	3	0.0	17.7	59.4

Benchmarks and metrics. Grounding is referring-expression comprehension on RefCOCO (1k frozen validation subset for development; full official testA/testB, $n = 5,657/5,095$, for confirmation) and held-out open-world detection on ODinW-13, which is never used for calibration, profiling, or any selection. General capability is VQAv2 (5k) and POPE (9k). Boxes are parsed from decoded text; unparseable output counts as wrong and the parse-failure rate is reported separately, since quantization can break the output format itself rather than the localization. Alongside the conventional REC@0.5 and mean GIoU we report *precise IoU*, the mean accuracy over thresholds 0.50:0.05:0.95, which is sensitive to box quality rather than mere overlap. All comparisons on full splits use a paired bootstrap clustered by image (10,000 resamples), since one image contributes several expressions.

The Hidden Failure Mode

Aggregate benchmarks report a near-lossless model. Table 1 gives the picture a practitioner would see. At W4, POPE moves 0.6 points and VQAv2 3.1; a model card reporting those numbers would describe the checkpoint as essentially intact. Correcting for the different chance floors of each benchmark (measured with an image-blind pass: REC 3.0, VQAv2 36.8, POPE 50.0), retention is 98.4% for POPE, 94.9% for grounding, and 92.8% for VQA—no dramatic asymmetry in the aggregate.

The metric decides whether the damage is visible. The aggregate view hides the shape of the damage. On the full RefCOCO testA split, W4 costs -3.78 points of REC@0.5 (95% CI $[-4.62, -2.95]$), -5.58 of mean GIoU $[-6.34, -4.83]$, and -6.64 of precise IoU $[-7.45, -5.82]$; testB gives -4.79 , -5.88 , and -7.08 with comparable intervals. Quantization does not primarily make the model point at the wrong object—it makes the boxes it draws less exact, which a threshold at IoU 0.5 is constructed not to notice. Whether an evaluator sees a 3.8-point or a 7.1-point regression is decided entirely by the metric they chose before looking (Figure 1a).

Held-out detection degrades further, concentrated on small objects. Transfer to open-world detection is harmed more than in-domain grounding: uniform W4 loses 11% of

relative ODinW F1 against 5% of in-domain REC. Stratifying held-out detections by ground-truth object size shows the loss is not uniform—recall on the smallest objects falls by roughly half while recall on large objects is unchanged, so an evaluation dominated by large, prominent objects registers almost nothing.¹ This is also why RefCOCO alone is a poor detector of the failure: its referring expressions target prominent objects, and its own area quartiles show no comparable gradient.

At 3 bits, the capabilities fail separately. Pushed further, the model does not degrade uniformly—it loses grounding first. Under GPTQ at 3 bits, grounding output is *exactly* zero: no parseable box on either RefCOCO or held-out detection, while VQAv2 retains 17.7 and POPE 59.4, both above their chance floors and verified to be varied rather than a degenerate yes-bias. Under RTN the same width destroys the model globally. Protecting the grounding band (below) at 3 bits more than doubles VQA ($17.7 \rightarrow 45.1$) yet still yields zero boxes: grounding is not only the first capability to fail, it is the most expensive to restore.

The Vulnerability Is Localized

To ask which weights carry the lost capability, we profile in the direction a deployment actually spends bits. Starting from the *fully quantized* model, we promote one decoder module m at a time (attention and MLP per block, ~ 56 modules) back to 8 bits and score the reduction in grounding damage on a held-out probe set with a decoding-free proxy,

$$s_m = \text{KL}_{\text{base}} - \text{KL}_{+m}, \quad (1)$$

where each KL is accumulated against the BF16 reference over exactly the coordinate-token positions of a teacher-forced target box. The probe set is drawn from training images and excludes every image appearing in any evaluation split.

Three properties of the resulting map matter for reliability (Figure 1b). It is **concentrated**: sensitivity peaks in a contiguous band of attention modules in layers 10–17 of 28, and MLP modules lose every per-byte comparison at our budget—converging with mechanistic evidence that localization is carried by a small number of attention heads in specific layer ranges (Kang et al. 2025; Schaumlöffel, Vilas, and Roig 2026). It is **capability-specific**: the same procedure run with a VQA probe yields a statistically unrelated ranking (Spearman $\rho = 0.001$; top-7 overlap 0/7), and the most grounding-critical module is VQA-negative—so protecting general capability and protecting grounding are different interventions, and an allocator tuned on VQA will not find these weights. It is **quantizer-specific**: maps profiled under RTN and under GPTQ correlate at only $\rho = 0.082$, and transplanting the RTN map onto a GPTQ checkpoint costs 1.2 points, while the matched map restores parity. An audit must therefore profile the quantizer that will actually be deployed.

¹The size-stratified analysis is being recomputed on the complete ODinW test split with predeclared relative-area quartiles and clustered intervals; the qualitative conclusion (small objects absorb the loss, large objects do not) is what we rely on here.

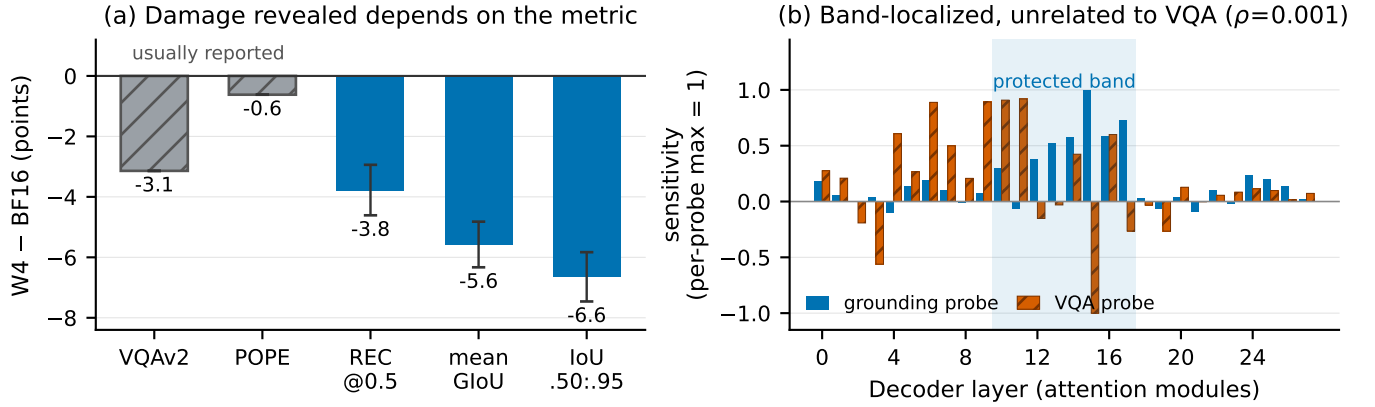


Figure 1: **(a)** How much 4-bit damage each metric reveals on full RefCOCO testA (bars are absolute point changes; whiskers are image-clustered 95% intervals). The benchmarks usually reported on model cards (hatched) show almost nothing; within grounding, a precision-sensitive IoU average reveals $1.8\times$ the loss that the conventional REC@0.5 reports. **(b)** Per-layer attention sensitivity from the promote-one-module probe, each probe normalized by its own maximum. Grounding sensitivity concentrates in layers 10–17; the VQA ranking is unrelated ($\rho = 0.001$), and the single most grounding-critical module is *negative* for VQA.

Table 2: Allocation policies at matched memory (avg. 4.25 bits: uniform W4 plus ~ 88 M parameters promoted to 8 bits, +6% decoder weight memory), on the frozen 1k RefCOCO validation subset used for all development decisions. Only grounding-guided allocation converts the same bytes into grounding; full-split confirmations with intervals are given in the text.

Policy	REC@0.5	GIoU	VQAv2
Uniform W4	84.5	0.754	78.0
Random promotion (3 seeds)	84.9 ± 0.3	0.758	78.2
VQA-guided promotion	85.1	0.764	78.6
Grounding-guided (ours)	86.6	0.775	78.4
at 4.5 bits	87.3	0.781	78.7

Mitigation at Matched Memory

Given the map, mitigation is deliberately the simplest possible intervention: greedily promote modules by sensitivity per byte to 8 bits under an average-bit budget, over an otherwise untouched quantization pipeline. Nothing is retrained and the calibration set is unchanged, so any gain is attributable to *where* the bits go.

Table 2 isolates that attribution. At an identical byte budget, random promotion buys +0.4 points and VQA-guided promotion +0.6, while grounding-guided promotion buys +2.1. On the full official splits the gain is significant under every metric and, consistent with the failure being one of precision, is largest under the strictest: testA REC@0.5 +1.84 (95% CI [1.24, 2.45]), GIoU +2.39 [1.85, 2.93], precise IoU +3.02 [2.44, 3.61]; testB +1.53 [0.78, 2.31], +2.15 [1.48, 2.83], +2.88 [2.20, 3.58]. The band selected on RefCOCO transfers to held-out detection, and general capability is not traded away: VQAv2 rises 0.4 points above the uniform baseline and POPE is unchanged at 88.8. Because the

intervention is training-free, this non-interference is structural rather than an empirical accident—there is no objective that could quietly barter chat quality for boxes.

Recovery saturates near two thirds of the loss by 4.75 bits, and a probe reweighted toward the smallest boxes selects a partly different allocation but does not improve small-object recovery. The residual is the honest limit of a training-free intervention, and motivates recovery training as a clearly separate second stage.

Discussion: What Reliability Requires

Our results argue that aggregate accuracy is the wrong certification for a compressed VLM, in three concrete ways. **Capability coverage:** a suite of VQA, hallucination, OCR, and reasoning benchmarks assigns a near-lossless verdict to a checkpoint that has lost a measurable fraction of its spatial precision and roughly half of its small-object recall. **Metric choice:** even when grounding is evaluated, a threshold at IoU 0.5 understates the regression by $1.5\text{--}1.8\times$ relative to a precision-sensitive average; strict-IoU reporting should be the default for compressed models. **Operating-point coverage:** the capability ordering at 4 bits does not predict the ordering at 3 bits, where grounding fails while language survives, so a single tested width does not certify a family of checkpoints. Concretely, we suggest that compressed-VLM releases report grounding at strict IoU, stratified by object size, alongside their language benchmarks—and that toolchains profile sensitivity on the quantizer they deploy, since the maps do not transfer.

Limitations. This is a single-model study; the specific band is a property of Qwen3-VL-2B and the generality of its location is untested. Our RTN baseline is deliberately plain, so the 3-bit collapse is a property of that baseline as much as of the width. The sensitivity proxy is teacher-forced and per-module, so it is used only to order a search—every configu-

ration reported here is selected on decoded metrics—and its per-module validation is underpowered on its own. Finally, the size-stratified detection analysis is being recomputed on the full held-out split with predeclared quartiles, as noted above.

Conclusion

Quantization does not damage a vision-language model uniformly, and the benchmarks that certify compressed checkpoints are the ones least able to see the damage it does. On Qwen3-VL-2B, 4-bit weights leave hallucination and question answering nearly intact while degrading spatial precision by up to 7 points under a strict criterion and roughly halving small-object recall; the vulnerable weights form a narrow, capability-specific, quantizer-specific band; and protecting that band recovers a substantial share of the loss for 6% extra memory with no training. The mitigation is useful, but the measurement is the point: reliability for compressed multimodal models has to be certified capability by capability, at the operating point and under the metric that can actually reveal failure.

Reproducibility. The evaluation harness, sensitivity maps, promotion lists, and frozen evaluation subsets will be released.

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